

UNDERDETERMINATION

THIS is where Chapters 3 and 4 have been leading. In the terminology of internal and external virtues, external virtues do not determine that a theory is true or that it is false. A theory can be explanatory and can be favorably tested against the observable evidence yet still be false. The logical structure of explanation and testing is never of the form that the evidence entails a theory. Rather, it is the other way around. A theory entails its evidence, and because of this, observation of the evidence cannot stand as clear confirmation of the theory. Nor can the evidence provide unambiguous falsification since, as we have seen, realistic cases of theory testing are of the form, theory *plus auxiliaries* entail the evidence, thus confusing the issue of assigning guilt in the case where the predicted event does not occur.

To put it bluntly, the observations of the world, that is, the reports on the manifest phenomena, are not sufficiently informative to single out the one true theory of what is going on behind the appearances. This is not to say that anything goes, that one theory is as good as another, or that observations have no significant role to play in the scientific justification process. Explanation and testing are strict and crucial discriminators of scientific theories in that a theory is unacceptable if it is in no way explanatory and does not cooperate with other theories to make true predictions. The comparison of a theory with observations weeds out lots of potential theories. The problem of underdetermination is that this process doesn't weed out enough, since many competing theories can account equally well for the evidence. The evidence alone

cannot trim the pool of applicants down to the single theory that is the true theory.

A theory with a perfect score on the evidence, in other words, is not necessarily true. It could explain all relevant phenomena, leaving no outstanding puzzles, and it could be confirmed by all tests and still be false. There will always be an alternative theory that scores just as well on the evidence, an alternative that is empirically equivalent. This other theory explains the same phenomena and makes the same predictions but does so with a different description of what is happening in the unobservable world. At most, one of these alternatives is right about how the world works, and insofar as we are interested in this basic understanding, we have a stake in determining which one it is.

It is not necessary for us to be aware of the alternative theory or for scientists actually to use it to generate predictions for it to upset the process of justification. There are plenty of situations in science where only one theory is imaginable. But this is just a limitation on imagination, and as long as an alternative account is conceptually possible (and it always is), the observational data alone can give the theory we have in mind at best a fifty-fifty chance of being true. In fact, though, the problem of underdetermination is much worse than this because, for any finite amount of data (things explained, tests passed), there will always be *any number* of theories that are empirically equivalent over that data. Of course, the huge majority of those alternative theories are really dumb. Shy, moonstruck sirens rising from the oceans to cause the tides? Old, crusty rubber being scraped from the balloon to make it sticky? There is no doubt that these are silly theories, and no one in his right mind advocates them as being true. I don't and neither should you, but why? They are dumb theories as judged by internal standards. For example, they are far-fetched and wildly implausible, which is to say they violate all standards of entrenchment. The theory of sirens at least is unnecessarily baroque in its ornamental descriptions of humanlike creatures with

emotional lives and intentional activities. Plain old gravity will do just as well. Plain old gravity, that is, is simpler in its explanation of the same phenomenon. So it's not the evidence that is decisive between these rival theories. It can't be; the crazy alternatives are empirically indistinguishable from the real theories (or if they aren't, then we'll find some that are).

If there is any number of empirically equivalent alternatives, then the observational evidence alone cannot give any particular theory even a fifty-fifty chance of being true. The pool of empirically adequate theories is as large as you like, and it can be trimmed toward the truth only in cooperation with internal standards. It is crucial, therefore, to be sure of the truth-conduciveness of the internal standards.

As a graphic example of underdetermination, recall the simple experiment of Chapter 2 in which the length of a metal bar was measured as the bar was heated. This example is far down at the simple end of the spectrum of complexity of scientific activities, and so it is not fairly characteristic of the complication of experiments or of the general difficulty in discovering alternative, empirically equivalent theories. It is, nonetheless, generally descriptive of the information imbalance in every case where theories meet the world. There is a finite amount of data, and this is always the case. All the past observations of things to explain, plus all the future tests to be made, sum to a large but finite number of data points. The number of theories that can successfully explain what wants explaining and that pass the tests, on the other hand, is unlimited. Among these empirically indistinguishable theories will be a variety of alternative descriptions of how the unobservable world works.

The thermal-expansion example is such a good representation of the imbalance, not because it is a particularly underdetermined experiment but because it is particularly amenable to graphic presentation and because the key moves in the explanation and confirmation activities are clearly identifiable. Figure

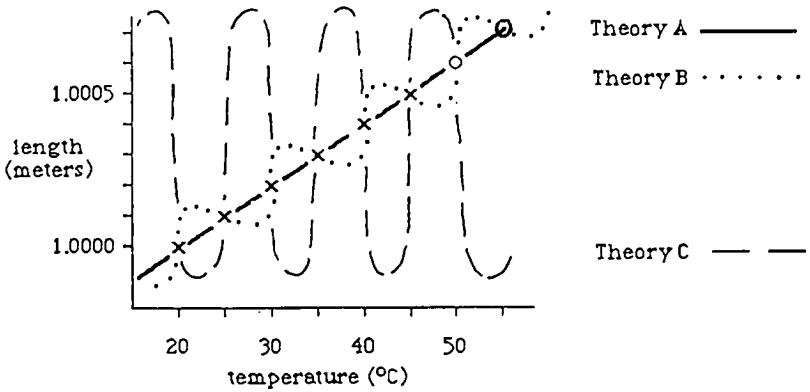


Figure 5.1. The heated-bar example (a reproduction of Figure 2.2).

5.1 repeats the graph of the data given in Chapter 2. The past data, things that have already been measured, are displayed on the graph as crosses, and the future data, outcomes of future tests, are shown as circles. A theory of what is going on here, that is, of the mechanism by which heating the bar causes it to change size, will describe what length the bar *would* take at any particular temperature. The theory is descriptive of hypothetical cases as well as actual cases. It goes beyond the actual evidence; that's what makes it interesting and useful. In this case a theory will provide a continuous line on the graph.

The theory represented by line A explains all the observed events and correctly predicts the future test at 50 degrees. Graphically, A passes through all the data points. But then, so does B. It is easy to see that for any finite number of data points there will be plenty of curves to choose from that pass through all the points. There will be an unlimited supply of empirically indistinguishable theories, but this does not mean that the scientist should advocate anything but theory A. It is foolish to conjure up curves in the account where none are needed. Theory A is preferable on standards of simplicity, and this shows off the importance of the internal feature of the theories.

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The experiments of measuring the length and temperature of the bar demonstrate that the world *seems* as if theory *A* is true. There is no doubt that things appear this way. But, based on the observations, it appears as well as if *B* is true, and clearly both cannot be right. This forces a close look at the question of the inference from evidence to theory, the inference from “seems as if” to “is.” Even when no alternative theory *B* has been imagined, there is reason to doubt the move from the evidence that the world seems as if *A* to the claim that the world really is *A*.

What is the responsible thing to do in the face of this doubt? One solution is simply to avoid the inference altogether by precluding science from making claims as to how the unobservable world *is*. Science, from this perspective, is in the functional business of reporting what the world seems like. It still deals in theories about unobservables, claiming with conviction that, by all experimental accounts, events in the world behave *as if* there are atoms (just as the events each Sunday morning are *as if* my neighbor steals my newspaper). Scientists are committed to the truth of such “as-if” claims, but this does not force a commitment to the corresponding “is” claim. Science, by this view, does not claim that the brass bar really expands in a linear response to rise in temperature, only that the bar is *as if* this is the case, just as it is as if the oscillation theory *B* is the case. What counts in science is finding an empirically adequate theory, and *A*, *B*, or *C* will do just as well. Truth about the unobservable is beside the point. It doesn’t matter which of the empirically indistinguishable theories *A*, *B*, . . . , is adopted because they are only valuable for empirical purposes of explaining and predicting. For these purposes the alternatives are equally adequate, and there is no pressing decision to be made in choosing one over the others.

This response to underdetermination, diminishing the aims and responsible accomplishments of science in proportion to the limited informational content of experience, comes in two basic varieties. There are interesting variations on each of the two

approaches, but these two perspectives will sufficiently broaden our view of the issue. One school of thought has the attitude that science doesn't (and shouldn't) intend theories to be thought of as being true or false. The true-or-false property is not a feature of theories. Theories are useful or not, applicable or not, but they are not true or false. This view of theories as useful, intellectual tools is often called **instrumentalism**.

Alternatively, theories might be seen as genuinely true or false, but underdetermination may prevent our ever knowing which score a particular theory deserved, true or false. When scientists theorize about atoms they really mean to be talking about real things. They *intend* to be telling the truth, but in light of underdetermination, the responsible thing to do is to avoid committing oneself one way or the other. The theory is true or false, but we cannot tell which. This is the position of the **empiricist**.

These two views of science and the status of theories are two varieties of antirealism. This is an umbrella organization defined by its opposition to **realism**, a view that the theories of science are in fact true or false of the world, or at least approximately so, and that we can sometimes tell of a theory whether it is true or false. In some cases, according to the realist, the inference from "seems as if" to "is" is justified. Realism clearly bears the burden of proof to show just how this inference is to work and to specify for what kinds of cases the inference is justifiable. Realism has the responsibility to deal with underdetermination. To see how this burden might be carried, we should look closely at the two versions of antirealism and the most persuasive realist response to each.

THE RESPONSIBLE AIMS OF SCIENCE

The key to instrumentalism is the focus on the proper intent behind the use of a scientific theory and on the appropriate question to ask in evaluating theories. The proper question to ask about a

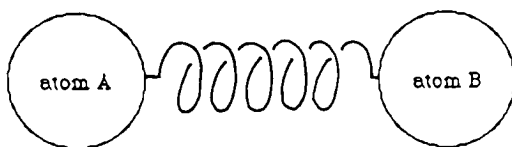


Figure 5.2. A model of the diatomic molecule AB , with the binding force represented as a spring.

theory is not whether it is true or false, because theories simply do not have this property, just as a hammer doesn't have the property of being true or false. A hammer, like any other tool, is judged for its usefulness and its applicability to a particular task. A hammer is more useful than a potato, for example, for driving nails.

Theories too are tools, by the instrumentalist account. They are intellectual tools for purposes of prediction of future phenomena and organization of observations. This is their intended role in science, and so the relevant criteria of evaluation of scientific theories are criteria of service, of usefulness and applicability rather than truth. Empirical adequacy – efficient organization and successful prediction of phenomena – is the goal of theorizing. There is no intent, indeed, no reason, in the mind of the instrumentalist, to push theories beyond their means by attempting to use empirical adequacy as evidence for truth. Equally empirically adequate theories are equally acceptable. As there is no one true hammer, there is no one true theory of gravity.

On this view, theories function as pedagogical tools for experts. Teachers of science will often invent intellectual devices to help their students understand the physical world. It is not uncommon in an elementary chemistry class, for example, to be given license to picture the atoms in a molecule as being connected by inner springs (Figure 5.2). This is a harmless aid to understanding the basics of physical chemistry. If you think of there being tiny springs between the atoms, you'll get the calculations and

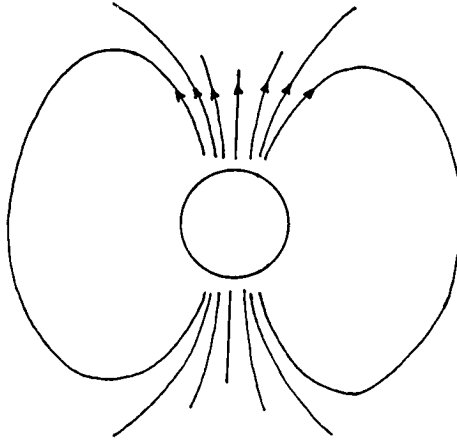


Figure 5.3. Magnetic field lines of the earth's magnetic field.

predictions right, and a lot of what molecules do, things like their oscillations, specific heat, and optical properties, will make sense. It is a very useful way to picture the molecules, but no chemistry teacher means to say that there really *are* tiny springs between the atoms. The claim about the real world is only that it is *as if* there are such springs.

Experts, not just students, use these sorts of conceptual aids. The magnetic field of an object, the earth for example, is often pictured and thought of in terms of the magnetic field lines (Figure 5.3). These lines are incredibly useful. Rotatable magnets like the needle of a compass will line up tangent to a field line. The density of lines, the number per volume, is related to the strength of the magnetic field, and an electric current is produced in a loop of wire if the number of lines that pass through the loop changes over time. Physicists often speak of “cutting field lines” to describe passing the wire loop in and out of the magnetic field. But it would be wrong to think that just because you can cut them they are real. No one even suggests that there really are lines around a magnet or around the earth. The references to magnetic

field lines are not intended to be taken literally, and so it would be inappropriate to call them true or to call them false. Claims about the field lines are simply useful. The description of field lines is an effective intellectual tool for dealing with magnetic phenomena, neither more nor less.

This interpretation of the status of scientific theories applies generally to all theories, according to the instrumentalist. Science has demonstrated very nicely that the world is as if there are atoms, tectonic plates, genes, a Minoan civilization, and so on, and that's enough. That is the proper goal of science, and with empirical adequacy the job is done. There is no question of whether the theories are true or false, of whether atoms really exist, because the theories are not meant to be true or false. Quarks are just like inner springs, only for grown-ups.

This view of the proper aims of science, the disavowal of any claim to a true account of the unobservable, is motivated by a desire to distance science from the occult. If appeals to mystical, unseen forces as in astrology or psychokinesis are to be disregarded as meaningless mumbo-jumbo, then one has to wonder how science is any different (that is, any better) if it invokes mystical, unseen entities such as electrons and genes. Instead of answering the question, the instrumentalist disarms it. The worry is dissolved if science simply stops talking about such things, or at least stops endorsing talk about unobservables like electrons and genes as true.

The instrumentalist move buys safety for the scientific enterprise, but it pays with interest. Instrumentalist science is no longer interesting science. It gives up what is most worth doing in science, namely, understanding what is happening behind the scenes in the realm of unobservables. The world, both those features that are manifest and those removed from experience, is in fact one way or another. Something causes the world as we see it. Observable objects are constituted of something. The past was

one way or another. And in all these cases we want to know about it. This kind of knowledge, contrary to instrumentalism, is and should be the goal of theorizing.

Instrumentalism, in other words, asks too much of us in that it asks us to ignore our curiosity about the unobservable. More is expected of science than simply to drift in the shallows of empirical adequacy.

Of course, this tough talk is cheap. The imposing question is whether the interesting aspect of science, real descriptions of the unobservable world, can be recovered without undue loss of security. If theories are intended as more than pedagogical tools and calculating aids, if theories are supposed to be true or false, does science have the ability to distinguish the true from the false?

RESPONSIBLE ABILITIES OF SCIENCE

Grant science its aims of producing true accounts of both the observable phenomena and the unobservable causes and constituents. The cautious empiricist will do this but will also point out the problems of underdetermination. All confirmation, that is, all distinctions of the likely-to-be-true from the likely-to-be-false theories, will be seriously equivocal, and testing against the evidence cannot distinguish any one of the unlimited number of empirically equivalent theories. The empiricist will also point out the problems with explanation. Explanation in general, judged by its contribution to our understanding of things, is a pragmatic and psychological event with no clearly truth-conducive features. If we specialize to strictly causal explanation, the evidence still underdetermines which causal explanation is the right causal explanation. The extra kick from a causal account, recall, is what you get once you know that you are dealing with the *correct* causal explanation. To paraphrase Nancy Cartwright, if God tells us (that is, we are certain) that nuclear fusion *explains* the sun's energy, we can accept this and yet have no reason to believe that

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nuclear fusion is an actual solar process. But if God tells us that nuclear fusion *causes* the sun's energy, then we know as well that the nuclear fusion really takes place. This is good news as far as it goes, but of course God doesn't go around saying these things, nor does the natural evidence speak in such unequivocal terms of the correct causal explanation.

Given this underdetermination of theory by evidence, responsibility demands an attitude of agnosticism toward unobservables. Theories, according to a dominant group of empiricists, are indeed to be construed as true-or-false about the unobservable world, but it is most appropriate, by these empiricist standards, to withhold belief in the existence (or not) of unobservable entities and in the truth (or not) of theories about unobservables. The idea is not to say that there are no such things as quarks, for example. It is rather to say that we simply don't know and we in fact *can't* know if there are such things. The theory is in fact true or false, but we can't tell which. A claim about quarks or atoms, for example, is beyond the limits of justification and so beyond the limits of knowledge. Though there is good reason to use the theory *as if* it is true, there is no good reason to believe that it *is* true.

The description of this position as agnosticism is entirely appropriate because there is a significant similarity between questions of the existence of unobservable scientific entities and the question of the existence of God. We can ask whether talk of God is literal or metaphorical. Is reference to God merely a metaphorical way of talking about nature and the amazing (as well as the regular) things that happen in the world? Is talk of God just a way of dealing with otherwise overwhelming issues of life, death, and mortality? If we think *as if* there is a designer of our existence and a judge of our actions, then a lot of things make sense and there is purpose to living. Granting this, an atheist might claim that if there must be talk of God it should be construed as instrumental talk. There is no such thing as God, just as there are

no springs attached to the atoms and no lines of magnetic force around the earth. If you must think in these terms to organize your affairs, that's fine. Just realize that it is only a useful invention you refer to as God, field line, or quark. In this sense, there are similarities between atheism and instrumentalism.

The more cautious agnostic would point out that, although God's existence cannot be proved, neither can God's nonexistence. There is no clear evidence that there is or that there is not such an entity. Therefore, either claim, of the atheist or of the believer, is unjustified, and in fact unjustifiable. The issue is beyond the limits of knowledge, and the appropriate attitude, the responsible attitude, is to withhold judgment. Thus the religious agnostic shares standards of justification with the scientific empiricist.

This parallel between reasoning in science and in religion is something to keep an eye on. God and electrons are both unobservable. Can we justify claims about one but not about the other? If we can prove that electrons exist, will the same style of proof demonstrate the existence of God? Why not?

The issue of scientific realism pivots on the distinction between empirical adequacy of a theory and the truth of a theory. Clearly these are distinct features. They mean different things. A realist view of science, though, sees the two as linked and must accept the burden to demonstrate this link. It's not that empirical adequacy (something that is directly testable) is simply the criterion of truth (a feature that is not directly testable), that is, that empirical adequacy is all there is to truth. The realist claims rather that empirical adequacy is a symptom of truth, meaning that compatibility with observations is an indication of a deep truth, truth about unobservables.

The standard presentation of the link between empirical adequacy and truth of a theory is known as the inference to the best explanation. Since it aims to exploit the truth-conducive virtues of explanation it is most effectively applied to causal explanation. Simply put, the realist claim is that there is good reason to believe

in the truth of the theory that is the best causal explanation of the phenomena. Identifying the *best* explanation limits the collection of acceptable theories to just one, and insisting on a causal explanation insures that the theory is not explanatory only for its pragmatic qualities. The causal claim, recall, is definite in what it says is true of the unobservable world. An entity must actually exist to be a cause. It is not enough for it simply to *seem* to exist.

The realist position is that it is possible to justify some claims about unobservables by noting whether those claims function in the best explanation of observable phenomena. Being unobservable, in other words, does not automatically put something beyond the limits of knowledge. This is not an across-the-board endorsement of the belief in the unobservable entities of science. Some claims about unobservables are warranted and some are not. Furthermore, the justification for belief in a theory can change. Newtonian mechanics is a perfect example. For a long time the theory's explanatory and predictive success and its internal plausibility provided justification to believe the theory to be generally true. But both internal and external evaluations change, and the theory is now less likely to be true. It's not that its truth value changes. Either it is true or it's false for all time. What changes is our warrant to believe one way or the other. Under the circumstances, it was once justifiable to believe the theory. Circumstances change, and so there is no guarantee that what is justified now will always be. But you do the best you can; that's the responsibility of science.

The realist's cause, that inference to the best explanation is legitimately inference to the likely truth of the theory, hinges on the notion of being the *best* explanation. To license the inference, this bestness must be an evaluable feature and not one of these "if God tells you it's the best" arrangements. And there must be an explicit demonstration that being the best, in whatever sense defined, is a truth-conducive feature. Clearly, the standards of being the best will be largely, if not entirely, internal. Being an

explanation is an external virtue since it involves comparison with observation, but being the best among empirically equivalent explanations will be a matter of being a simple, precise, and fruitful theory that is compatible and cooperative with other acceptable theories. Determining the best explanation will demand trading off some internal virtues for others and finding the optimum balance.

None but the most reckless realist would say that the best explanation is guaranteed to be true. The inference concludes only that the causal explanation that is most fruitful, that is, explains the most phenomena with the minimum of theoretical extravagance, and that fits into the larger theoretical network is more likely to be true than a theory that does none of this or does only some of it. It is unlikely that a false but adequate guess at an explanation of one phenomenon will both fit in with other scientific theories and contribute to explanations of other phenomena and other theories. The demand for coherence among theories and the goal of maximal coherence limit the number of possible theoretical pictures of the world and limit it in the direction of truth. The antirealist's sharp restriction of knowledge at the line of unobservability ignores this aspect of justification and preempts even comparative evaluation of, among theories of unobservables, which are more likely to be true.

Inference to the best (causal) explanation, in its most intuitive appeal, is this: The explanatory and predictive success of theories would be an absolute miracle if the theories were not true. It would be a miracle, which seems to happen over and over again in science, for a fictional account to fit consistently onto the reality of observation. This appeal has particular force in cases where several theories coincide, that is, where coherence among theories is clear. Consider the early experiments to measure Avogadro's number, the number of molecules of substance contained in one mole. This number is of course based on a molecular theory. The world is *as if* there are molecules, but does that

indicate there really *are* molecules? In this case there is warrant to say yes. The number of molecules in a mole can be measured in a variety of ways, and predictions of the number can be generated from a variety of theories. It can be measured by chemical, thermodynamic, electrical, microscopic-motion (Brownian motion), and other techniques. All of these experiments come up with the same number. There is no collusion among these various theories; hence it would be a staggering miracle, a stroke of phenomenal luck, for them all to agree, unless they were measuring something real, namely, real molecules. There seems no accounting for this coherence among independently developed theories other than to give them a united basis in truth. If the molecules really exist and molecular theory speaks the truth, then agreement in counting molecules is to be expected.

This sort of reasoning is strongly analogous to methods used to translate unfamiliar languages for which no dictionary or lexicon is available. Scientists are in a predicament similar to breakers of codes or translators of ancient texts written in unfamiliar languages. On the basis of what appears, the printed words in the translator's case, the basic observations of phenomena in the scientist's, the task is to finesse the meaning of the story of what the world is like. An instrumentalist would claim that there simply is no single meaning of the book and one reading is as good as another, as long as it is consistent with the marks on the page, that is, with the observations. An empiricist of the sort we are following would say there is a single, real meaning. The plot is one way or another, but without meeting the author and without the definitive lexicon, we can never know what the meaning is. The realist, as characterized above, agrees that the book has a single, true meaning, and furthermore, the translator can hope to figure it out, just as science can figure out the single, true description of the way the world is. Some claims about the plot, some theories, will make more sense than others and will fit more easily and cooperatively with the claims about other aspects of the plot, and

under the assumption that the book makes sense as a whole, only theories that fit a coherent account are likely to be true. Of course there are multiple coherent readings and hence multiple possible meanings, but as more of the book is examined and more pieces of the plot come to the light of theorized interpretation and must be fit with each other, insistence on coherence can shrink this collection in the direction of truth.

THE SIGNIFICANCE OF OBSERVABILITY

The differences between realism and antirealism are largely over the significance of the distinction between observable and unobservable events and entities. It is a question of the significance of this distinction as a limit of responsible knowledge claims. The antirealist, specifically the empiricist, regards observability as significant, since we are justified in claims about observables but not in claims about unobservables. We can judge as true or false claims about stones or clouds or planets but not claims about quarks or cosmic rays. There is warrant to believe statements about the Parthenon but not about Pericles. The distinction is between two distinct levels of justification. We are justified (according to the antirealist view of science) to believe claims about observables, but for things like atomic theory and other claims about unobservables, there is at best justification to use the claim as if it is true.

The realist view of science sees no sharp limiting of justification by the factor of observability. Justification can cross the line of observability. We want to know the truth about unobservables just as we want to understand the plot of the book and not simply know the shape of the marks on the page, and so there is need at least to compare theories and rate them as more or less likely to be true. This need is filled by the inference to the better or, in the context of more than two options, the best explanation. The realist will further undermine the significance of observability by

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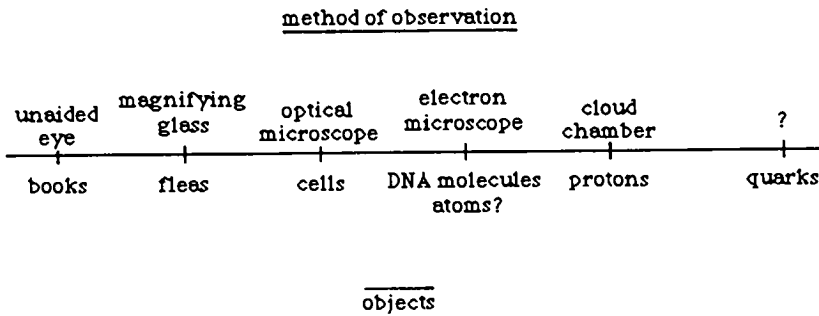


Figure 5.4. A spectrum of indirectness of observation, getting more indirect to the right.

pointing out that the line between what is observable and what is not is itself unclear. If instruments such as microscopes and cloud chambers do not count as direct observations of microbes and protons, then what do we say of things seen with the aid of a simple magnifying glass or eyeglasses or even a windowpane? What about observing things through the air? The point is not that there is *no* difference between evidence viewed with the unaided eye and evidence dependent on the use of a microscope or a cloud chamber; it is rather that there is a difference of degree. Entities do not fall neatly into one of the two categories, observable or not; they occupy a more or less continuous spectrum of indirectness of observation (Figure 5.4). The empiricist has the obligation to explain where on this spectrum is the sharp division, or even hazy division, that separates what we can know about from what we cannot. Admittedly, there are clear cases of observables and unobservables. We can see stones, but we cannot see quarks. But so many of the interesting entities of science will not be so clearly either observable or not, and it is the justification of theories about these things that we are interested in.

The empiricist and the realist both have good points, and there is nothing to prevent us from taking what is good from each and synthesizing our own account. Clearly, the warrant for belief in a

theory generally decreases as the observational evidence is less direct. The more apparatus necessary and the more inference used to sanction the observational claim, the more chance of misrepresenting the world. The empiricist is correct in saying that observability is significant to justification. I certainly have better justification for believing in the existence of this pencil in my hand than I do for believing in the carbon atoms in the graphite that is rubbing off onto the page, and more still than for believing in the quarks that constitute the atomic nuclei. But the justification for theories does not abruptly fall away at any point along the spectrum of observability. There is no sharp limit of knowledge where the justification meter goes all of a sudden from full to empty. In other words, the realist is correct in that the significance of observability does not force a yes-or-no dichotomy of justification.

Benefiting from the good parts of both realism and antirealism, the most profitable approach to understanding scientific knowledge and scientific method is to look for *degrees* of justification. Measurement of these degrees will be in terms of internal virtues and external comparisons to evidence, and it will even be sensitive to degrees of observability of the objects and events referred to by a theory. Observation is a funny concept. We might have thought that either something is observed or it isn't and hence the entity is either observable or not. But there is more variety in the event of observation than that.